M5Shark Owner Notes

Unofficial Version

Created as a replacement PDF for the M5-Shark repository

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Status: unofficial. This document is independently written and is not a factory manual, warranty document, compliance approval, or legal authorization. Use the device only for lawful education, repair, interoperability study, and authorized testing.

Purpose

This short owner guide gives a clean, safety-first starting point for the M5Shark hardware ecosystem. It is meant to help an owner identify the device, prepare a controlled workspace, choose cautious setup steps, and avoid common mistakes before experimenting with firmware, accessories, radio modules, NFC, RFID, IR, GPIO, or storage features.

First Use Checklist

1. Confirm the exact device name, board revision, and firmware screen before installing or changing anything.
2. Inspect the enclosure, display, buttons, USB-C port, battery area, antenna connector, and microSD slot.
3. Charge with a known-good cable and power source. Stop if the device becomes unusually hot, swollen, unstable, or damaged.
4. Insert only a clean, known-good microSD card if the workflow needs storage.
5. Attach only antennas and accessories that match the connector, voltage, and frequency range.
6. Start with receive-first, demo, logging, or owned-device workflows.

Responsible Use

The M5Shark and similar learning tools can involve RF, IR, NFC, RFID, Wi-Fi, BLE, and GPIO behavior. These features are useful for education and diagnostics, but they must stay inside lawful and authorized boundaries.

- Work only with devices, remotes, tags, cards, networks, and lab targets you own or have written permission to test.
- Do not attempt to bypass access controls, interfere with communications, disrupt systems, or test public infrastructure.
- Keep transmit-capable experiments inside a controlled lab setup and follow local radio rules.
- Document what you changed so you can return the device to a known state.

Safe Setup Flow

Identify	Record the exact model, revision, current firmware, storage state, and accessory list.
Prepare	Use a stable cable, charged battery, matched antenna, and clean workspace.
Update	Use only firmware intended for the exact hardware revision. Avoid random forks or images for similar-looking boards.
Test	Begin with basic menus, storage checks, screen controls, and receive-first examples.
Recover	Keep notes, backups, and the original firmware source if available before making deeper changes.

Feature Notes

RF and Antennas

Match the antenna to the band and connector. Never power radio hardware in a configuration that requires an antenna unless the hardware documentation says it is safe to do so. Avoid transmitting outside controlled, authorized environments.

NFC and RFID

Use only cards, tags, and systems you own or are authorized to evaluate. Treat unknown tags and access cards as private property. Keep training examples limited to lab tags and documented demos.

IR

IR learning is best tested with household devices you own, such as a spare remote and a noncritical receiver. Avoid sending commands to equipment where unexpected behavior could cause loss, damage, or safety risk.

GPIO and Accessories

Check voltage and pinout before connecting modules. A wrong pin, reversed polarity, or unsupported voltage can damage the device or accessory.

Troubleshooting

Device will not power on	Try a known-good USB-C cable and charger. Let the device charge, then check for heat or battery issues.
Screen is blank	Confirm power, brightness, firmware compatibility, and whether the device appears over USB.
SD card missing	Reformat a spare card with a common filesystem and test again. Back up any existing card first.
Accessory not detected	Recheck pinout, voltage, connector type, firmware support, and whether the module requires external power.
Firmware behaves oddly	Return to a known compatible build for the exact device revision. Avoid firmware made for a different board.

Owner Notes

- Keep a small change log with firmware version, date, accessories, and results.

- Photograph wiring before powering prototypes.
- Label microSD cards used for experiments.
- Separate learning files from backups.
- Stop testing when behavior is unexpected and return to a basic known-good setup.

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